

60V, 0.6A/1A/1.5A, Synchronous Step-Down LED Driver

1 FEATURES

- AEC-Q100 Grade 1 Qualified for automotive applications
- Wide V_{IN} Range: 3.5V to 60V
- Up to 0.6A/1A/1.5A Output Capability
- Integrated both High-Side and Low-Side Power MOSFETs
- Spread Spectrum Option for Low EMI Applications
- Frequency Options: 400kHz, 1MHz, 2.1MHz
- PWM Dimming
- Fault Indicator for LED Short, Open
- 0.2V Reference for LED Current Sense
- High Efficiency PFM Operation
- Near 100% Duty Cycle Operation for Low Drop Out Operation
- Accurate Peak and Valley Clamp
- Internal Compensation
- Accurate VIN UVLO Protection
- Over-Temperature Shutdown and Recovery
- Operating Junction Temperature -40°C to 150°C
- Thermally Enhanced SOIC-EP8 Package (4.9mm X 3.9mm)

2 APPLICATIONS

- Automotive Lighting
- Dimmable Interior Lights
- General LED Illumination

3 DESCRIPTION

The LN33X61Q1 is a high efficiency, compact, synchronous step-down LED driver employing a constant frequency, peak current mode control architecture with internal compensation. It operates from an input voltage from 3.5V to 60V and integrates both high-side and low-side power MOSFETs with up to 0.6A/1A/1.5A output capability. The nominal switching frequency is fixed on 400kHz, 1MHz or 2.1MHz. Automatic frequency foldback at light load improves efficiency.

The LN33X61Q1 provides multiple features such as precision enable, fault indicator and internal soft start. Full protection features include input UVLO, over temperature shutdown, cycle-by-cycle current limit, and short-circuit protection. The fault indicator outputs logic low signal if a fault condition occurs.

PWM dimming is achieved by applying an external PWM signal at the EN/DIM pin. The average LED current is proportionally scaled with PWM duty cycle.

Typical Application Diagram

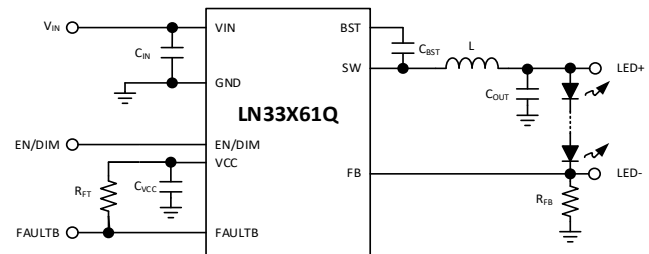


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4 REVISION HISTORY

Version	Change Description	Date
1.0	Initial Version	2022/01/26
1.1	Update Package Information.....P25	2022/04/26
1.2	Add new part number and corresponding specifications, update product and order information.....P4 Add parameters of “INTERNAL SOFT START” in EC Table.....P13 Add Chapter 12 “TAPE AND REEL INFORMATION”P27	2023/3/24

5 PRODUCT AND ORDER INFORMATION

5.1 Product Information

Part Number	Function	X	Y	IC Package	MSL-Peak-Temp ⁽¹⁾	Material	Top Marking ⁽³⁾
LN33061Q1YX	Fixed frequency, FAULTB	1: 1.5A, 400kHz 2: 1A, 400kHz 3: 0.6A, 400kHz 4: 1.5A, 1MHz 5: 1A, 1MHz 6: 0.6A, 1MHz 7: 1.5A, 2.1MHz 8: 1A, 2.1MHz 9: 0.6A, 2.1MHz D: 1.5A, 400kHz, Fast start-up ⁽⁴⁾	-: Normal S: SST ⁽²⁾	SOIC-EP8	Level-3-260C	RoHS	33061QYX
LN33161Q1YX	Adjustable frequency	1: 1.5A 2: 1A, 3: 0.6A D: 1.5A, Fast start-up ⁽⁴⁾	-: Normal S: SST ⁽²⁾	SOIC-EP8	Level-3-260C	RoHS	33161QYX

- (1) MSL (Moisture Sensitivity Level) and the highest solder temperature are based on JEDEC industrial standard.
- (2) Part numbers with **SPREAD SPECTRUM (SST)** function.
- (3) There may be additional marking relates to the internal trace code on the device.
- (4) Part numbers with **Fast Start-up** feature.

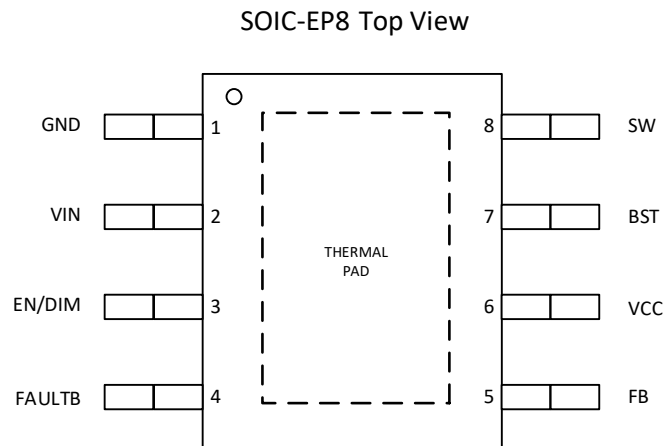
5.2 Order Information

Part Number	Output Current	Frequency	FAULTB	Spread Spectrum	Other Features	Package & Qty	Top Marking
LN33061Q1-1-EFR	1.5A	Fixed, 400kHz	Yes	No	Standard	Tape & Reel 3000	33061Q-1
LN33161Q1-1-EFR	1.5A	Adjustable	No	No	Standard	Tape & Reel 3000	33161Q-1
LN33061Q1-D-EFR	1.5A	Fixed, 400kHz	Yes	No	Fast start-up	Tape & Reel 3000	33061Q-D
LN33061Q1S1EFR	1.5A	Fixed, 400kHz	Yes	Yes	Standard	Tape & Reel 3000	33061QS1
LN33161Q1S1EFR	1.5A	Adjustable	No	Yes	Standard	Tape & Reel 3000	33161QS1
LN33061Q1SDEFR	1.5A	Fixed, 400kHz	Yes	Yes	Fast start-up	Tape & Reel 3000	33061QSD

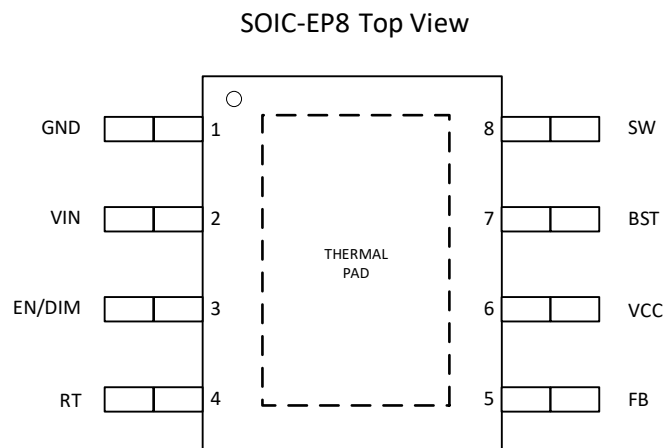
6 PIN CONFIGURATION

6.1 Pin Configuration

6.1.1 LN33061Q1 with FAULTB pin



6.1.2 LN33161Q1 with RT pin



6.2 Pin Functions

PIN Number	Name	Type	Description
1	GND	Ground	Power ground. Connect the pin to ground plane.
2	VIN	Power	Power supply input pin to high side power MOSFET and VIN LDO regulator. Decouple this pin to GND with ceramic capacitors.
3	EN/DIM	Signal	Enable pin for VCC LDOs, regulator output. Connect to VIN directly, or to VIN through a divider for VIN UVLO. This pin is also used for PWM dimming.
4	FAULTB (LN33061Q1)	Signal	Open-drain fault indicator. VFB is monitored and when VFB is not within the regulation window. FAULTB pin is pulled low.
4	RT (LN33161Q1)	Signal	Switching frequency control pin. Place a resistor between this pin and GND to set the switching frequency between 200kHz and 2.5MHz.
5	FB	Signal	LED current feedback input. Use an external current sense resistor to set the desired LED current.
6	VCC	Power	Voltage source that powers the gate drivers of the internal power MOSFETs and control circuits. Must be decoupled to GND with 1uF to 4.7uF ceramic capacitor. This voltage source is provided by one of the two internal LDO regulators with input from VIN or VOUT.
7	BST	Power	Bootstrapped supply to the high side gate driver. Connect a 0.1uF to 0.47uF ceramic capacitor between BST and SW pins.
8	SW	Power	Switch node connection from the internal power MOSFETs to the external inductor.
-	THERMAL PAD	-	Thermal dissipation pad. Solder to ground plane.

6.3 Package Thermal Parameters

Parameters ⁽¹⁾		SOIC-EP8	Units
$R_{\theta JA}$	Junction-to-Ambient Thermal Resistance	39	°C/W
ψ_{JT}	Junction-to-Top Characterization Parameter	6	°C/W

(1) Measurements are based on standard 2s2p PCB defined in JESD 51-7 2s2p, under no wind , 2W loss, and 25 °C ambient temperature.

7 SPECIFICATIONS

7.1 Absolute Maximum Ratings

Parameters	Min	Max	Unit
VIN to GND	-0.3	65	V
EN/DIM to GND	-0.3	V _{IN}	
VCC to GND	-0.3	5.5	
FB to GND	-0.3	5.5	
FAULTB to GND	-0.3	5.5	
RT to GND	-0.3	5.5	
SW to GND	-1.0	V _{IN} +0.3	
SW to GND (Overshoot voltage less than 10ns)	-3.5	65	
BST to SW	-0.3	5.5	
Ambient Temperature	-40	125	°C
Junction Temperature	-40	150	
Storage Temperature	-65	150	

7.2 ESD Ratings

Parameters	Min	Max	Unit
HBM Human Body Model		±3000	V
CDM Charge Device Model		±750	

7.3 Recommended Operating Condition

Parameters	Min	Max	Unit
V _{IN}	3.5	60	V
EN/DIM	-0.3	V _{IN}	
FB	-0.3	0.3	
FAULTB	-0.3	5	
I _{OUT} (1.5A version)	0	1.5	A
I _{OUT} (1.0A version)	0	1.0	
I _{OUT} (0.6A version)	0	0.6	
Ambient temperature	-40	125	°C
Junction temperature	-40	150	

7.4 Electrical Characteristics

Unless otherwise stated, the minimum and maximum limits are applied over the recommended operating junction temperature range of -40°C to 150°C. Typical values are measured at 25°C and represent the most likely norm. The default conditions are applied: $V_{IN} = 24V$, $V_{OUT} = 5V$, $F_S = 400kHz$.

Symbol	PARAMETERS	CONDITIONS	MIN	TYP	MAX	UNITS
SUPPLY VOLTAGE (VIN PIN)						
V_{IN}	Operating Input Voltage Range		3.5		60	V
$I_Q^{(1)}$	During Regulation	$V_{EN} = 5V$, $V_{IN} = 24V$, $I_{OUT} = 0A$		70		μA
I_{SHDN}	Shutdown Quiescent Current	$V_{EN} = 0V$, $V_{IN} = 24V$		1.2	3	μA
FB PIN						
V_{FB-CCM}	Regulated Feedback Voltage In CCM Mode	$V_{IN} = 3.5V$ to 60V, $T_J = 25^\circ C$	196	200	204	mV
		$V_{IN} = 3.5V$ to 60V, $T_J = 25^\circ C$ to 150 °C	192	200	208	mV
I_{Q-FB}	Feedback Input Leakage Current	$V_{FB} = 200mV$		0	100	nA
POWER MOSFETS ⁽²⁾						
$R_{DS(on)-HS}$	High Side MOSFET ON Resistance	$I_{OUT} = 1A$, $V_{BST} - V_{SW} = 5V$		340	650	m Ω
$R_{DS(on)-LS}$	Low Side MOSFET ON Resistance	$I_{OUT} = 1A$, $V_{CC} = 5V$		195	400	m Ω
OSCILLATOR ⁽³⁾						
$F_{S-400kHz}$	PWM Switching Frequency	$V_{IN} = 24V$, $V_{OUT} = 5V$	320	400	460	kHz
$F_{S-1000kHz}$	PWM Switching Frequency	$V_{IN} = 24V$, $V_{OUT} = 5V$		1000		kHz
$F_{S-2100kHz}$	PWM Switching Frequency	$V_{IN} = 12V$, $V_{OUT} = 5V$		2100		kHz
RT (RT PIN) ⁽⁴⁾						
$F_{S-RT-400kHz}$	PWM Switching Frequency	$V_{IN} = 24V$, $V_{OUT} = 5V$, $R_T = 232k\Omega$	320	400	460	kHz
$F_{S-RT-1000kHz}$	PWM Switching Frequency	$V_{IN} = 24V$, $V_{OUT} = 5V$, $R_T = 71.5k\Omega$		1000		kHz
$F_{S-RT-2500kHz}$	PWM Switching Frequency	$V_{IN} = 12V$, $V_{OUT} = 5V$, $R_T = 12.4k\Omega$		2500		kHz

(1) Guaranteed by design.

(2) Measured at pins.

(3) This parameter is for LN33061Q1YX with internal fixed frequency.

(4) This parameter is for LN33161Q1YX with externally adjustable frequency.

Electrical Characteristics (Continued)

Unless otherwise stated, the minimum and maximum limits are applied over the recommended operating junction temperature range of -40°C to 150°C. Typical values are measured at 25°C and represent the most likely norm. The default conditions are applied: $V_{IN} = 24V$, $V_{OUT} = 5V$, $F_S = 400kHz$.

PWM ⁽¹⁾						
T _{ON-MIN}	Minimum ON Time		150			ns
T _{OFF-MIN}	Minimum OFF Time		250			ns
SPREAD SPECTRUM (SST) ⁽⁵⁾						
%ΔF _S	Spread Spectrum Modulation Frequency Range		-18			%
F _{SST}	Spread Spectrum Modulation Frequency		16			kHz
ENABLE V _{IN} UVLO (EN PIN)						
EN _{VOUT-ON}	V _{IN} UVLO Rising Threshold	V _{EN} Rising	1.90	2.09	2.25	V
EN _{VOUT-HYS}	V _{IN} UVLO Hysteresis	V _{EN} Falling	240			mV
EN _{VCC-ON}	V _{EN} High-Level Threshold	V _{EN} Rising	1.8			V
EN _{VCC-OFF}	V _{EN} Low-Level Threshold	V _{EN} Falling	0.4			V
t _{EN-OFF}	EN Turn Off Delay		10	20	30	ms
I _{Q-EN}	EN/DIM Pin Current	V _{EN} = 3.3V	10 200			nA
INTERNAL VCC LDOS (VCC, V _{IN} , V _{OUT} PINS)						
V _{CC-TARGET}	V _{CC} Regulation Target		4.75			V
BOOTSTRAP (BST PIN)						
t _{ON-MAX}	Auto-Refresh Period		9			μs
THERMAL SHUTDOWN ⁽¹⁾						
OT	Thermal Shutdown Threshold		170			°C
OT _{HYS}	Thermal Shutdown Recovery Hysteresis		-10			°C

(5) This parameter is only available on part numbers with spread spectrum function.

Electrical Characteristics (Continued)

Unless otherwise stated, the minimum and maximum limits are applied over the recommended operating junction temperature range of -40°C to 150°C. Typical values are measured at 25°C and represent the most likely norm. The default conditions are applied: $V_{IN} = 24V$, $V_{OUT} = 5V$, $F_S = 400kHz$.

OVER CURRENT PROTECTION ^{(6) (7)}					
$I_{PEAK-1.5A}$	Peak Current Limit Threshold	$V_{IN} = 24V$, $V_{OUT} = 5V$	1.75	2.2	2.5 A
$I_{VALLEY-1.5A}$	Valley Current Limit Threshold	$V_{IN} = 24V$, $V_{OUT} = 5V$	1.15	1.6	1.95 A
$I_{PEAK-1.0A}$	Peak Current Limit Threshold	$V_{IN} = 24V$, $V_{OUT} = 5V$		1.6	A
$I_{VALLEY-1.0A}$	Valley Current Limit Threshold	$V_{IN} = 24V$, $V_{OUT} = 5V$		1.1	A
$I_{PEAK-0.6A}$	Peak Current Limit Threshold	$V_{IN} = 24V$, $V_{OUT} = 5V$		1.0	A
$I_{VALLEY-0.6A}$	Valley Current Limit Threshold	$V_{IN} = 24V$, $V_{OUT} = 5V$		0.65	A
FAULT THRESHOLDS (FAULTB and FB PINS) ⁽⁸⁾					
$\%V_{FT-OV}$	Fault Over-Voltage Rising Threshold	V_{FB} Ramping Up		140	%
$\%V_{FT-OV-HYS}$	Fault Over-Voltage Recovery Hysteresis	% Of FB voltage		20	%
$\%V_{FT-UV}$	Fault Under-Voltage Falling Threshold	V_{FB} Ramping Down		85	%
$\%V_{FT-UV-HYS}$	Fault Under-Voltage Recovery Hysteresis	% Of FB voltage		20	%
V_{FT-PD}	Fault Pull-Down Strength	$I_{FT} = 1mA$	65	105	150 mV
$t_{FT-RISING}$	Fault Flag Rising Delay			8	μs
$t_{FT-FALLING}$	Fault Flag Falling Delay			16	μs
t_{EN-FT}	EN/DIM to Fault Delay		70	130	200 μs
I_{Q-FT}	FT Leakage Current	$V_{FT} = 5V$		5	100 nA
INTERNAL SOFT START ⁽⁹⁾					
t_{SS-INT}	Internal Soft-Start Time	For P/N with Fast Start-up feature	2.9	5	ms
		For P/N without Fast Start-up	6.6	10	ms

(6) This current limit was tested as the internal comparator trigger point, the current limits measured in a closed loop application may be different.

(7) Subscripts indicate different output current capabilities in the product family.

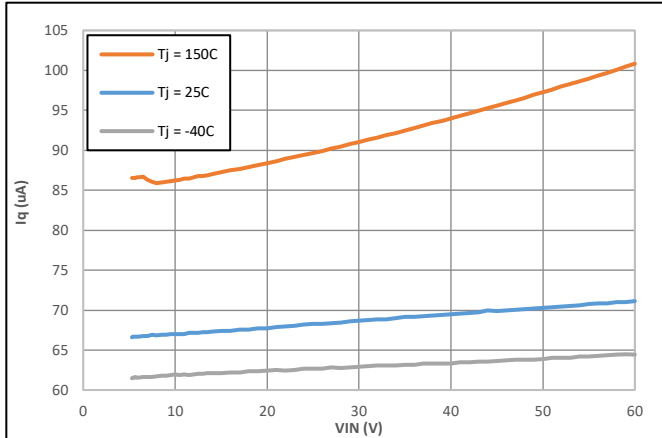
(8) Fault Thresholds are parameters of V_{FB} with respect to V_{FB-CCM} , these parameters are available on LN33061Q1 only

(9) Measured from $EN_{VOUT-ON}$ to internal soft start completed.

7.5 Typical Characteristics

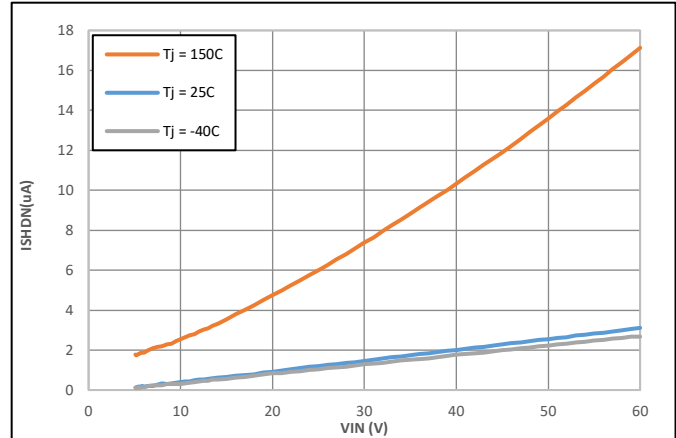
7.5.1 Characteristics Over Temperature

Unless otherwise stated, the test conditions are the same as Chapter 7.4. $T_J = -40^{\circ}\text{C}$ to 150°C .



$V_{EN} = V_{IN}$, No load

Figure 1. I_q



$V_{EN} = 0\text{V}$

Figure 2. I_{SHDN}

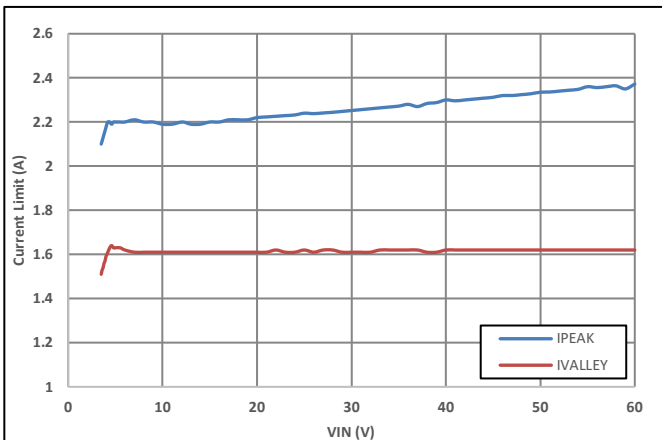


Figure 3. Peak Current and Valley Current Limit vs. V_{IN}

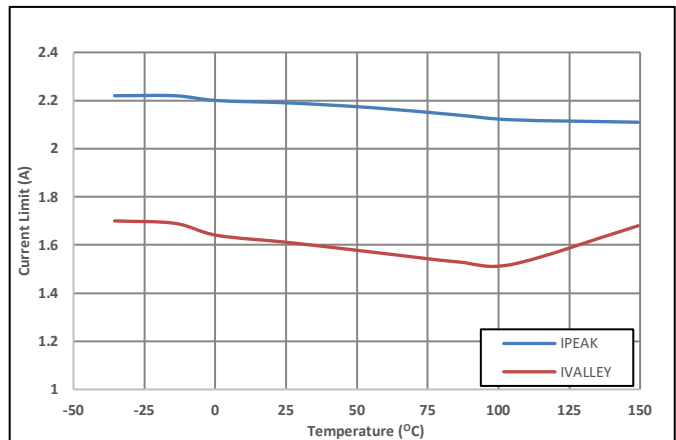


Figure 4. Peak Current and Valley Current Limit vs. Junction Temperature

7.5.2 Typical Characteristics

Unless otherwise stated, the test conditions are applied: $V_{IN} = 24V$, LOAD = 2 LEDs in series, $F_S = 400kHz$, $L = 15\mu H$, $C_{OUT} = 10\mu F$, $T_A = 25^\circ C$.

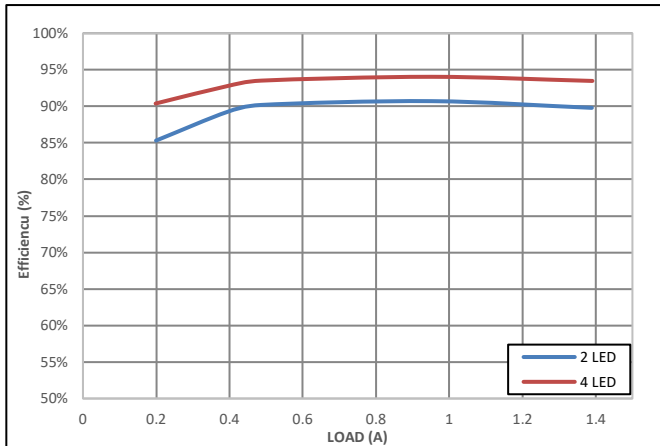


Figure 5. Efficiency Curves, $V_{IN} = 24V$

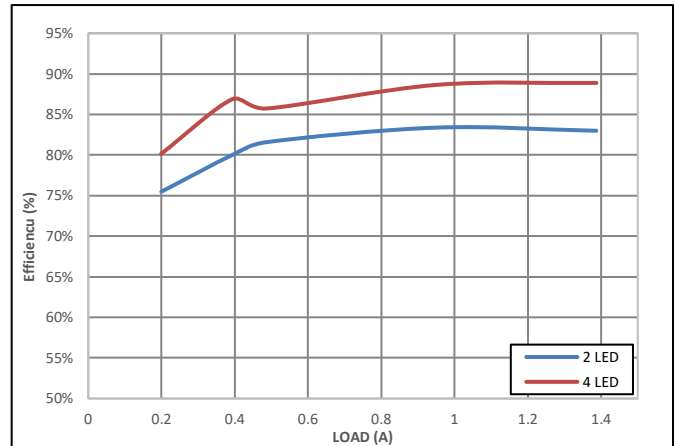


Figure 6. Efficiency Curves, $V_{IN} = 60V$

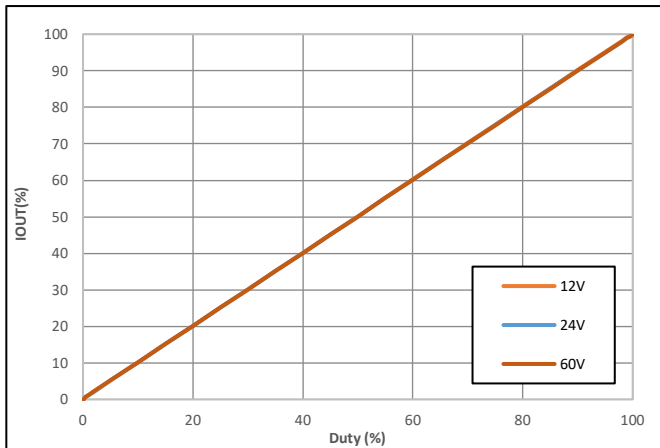


Figure 7. Dimming Curves, Dimming Frequency = 100Hz

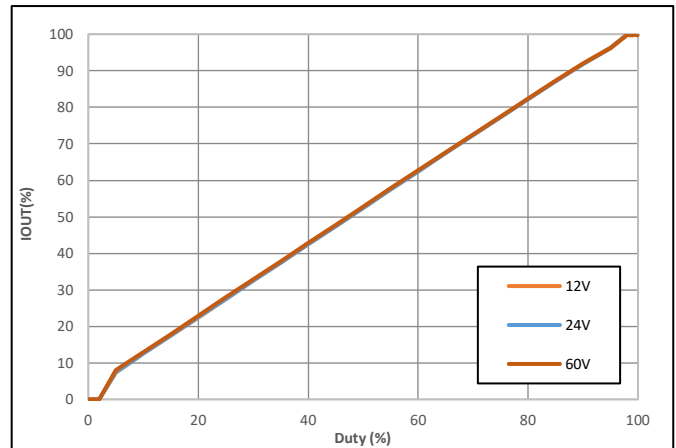
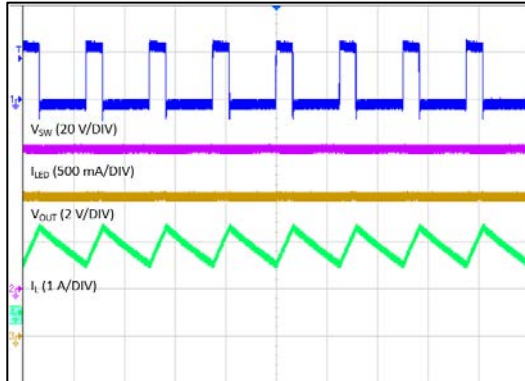


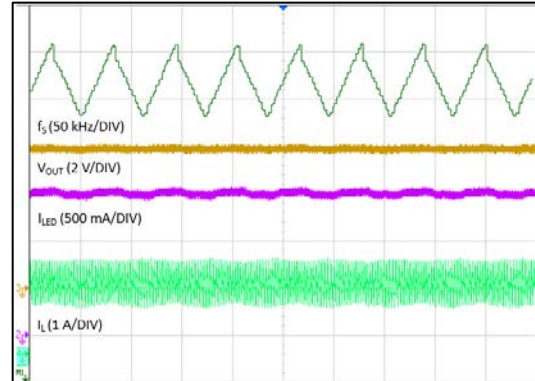
Figure 8. Dimming Curves, Dimming Frequency = 2kHz

7.5.3 Typical Waveforms

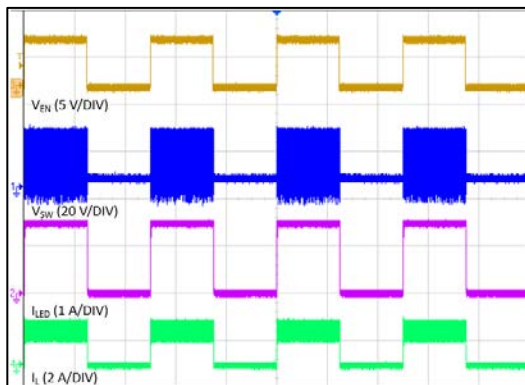
Unless otherwise stated, the test conditions are applied: $V_{IN} = 24V$, LOAD = 2 LEDs in series, $F_S = 400kHz$, $L = 15\mu H$, $C_{OUT} = 10\mu F$, $T_A = 25^\circ C$.



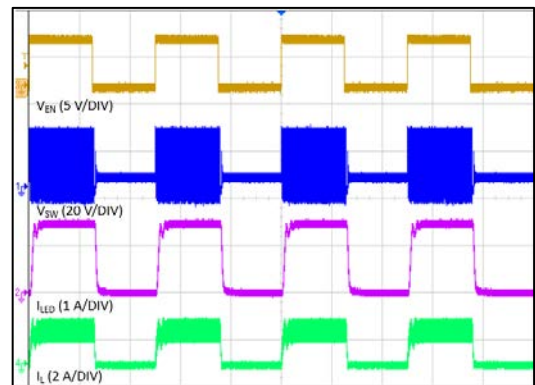
Time (2 μs /DIV)
Figure 9. Steady State, $I_{LED} = 1.5A$



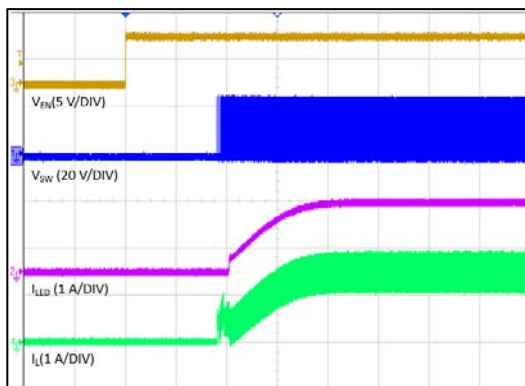
Time (50 μs /DIV)
Figure 10. Steady State, $I_{LED} = 1.5A$, Spread Spectrum



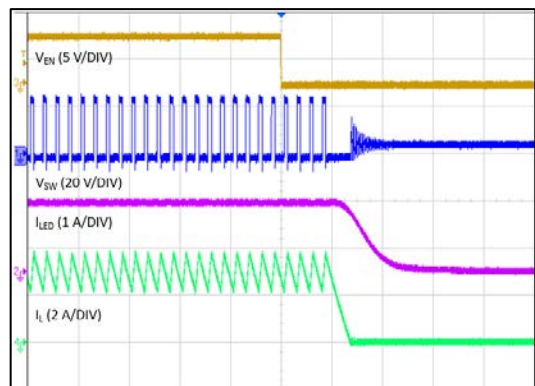
Time (2 ms/DIV)
Figure 11. PWM Dimming, 200Hz, 50% Duty Cycle



Time (200 μs /DIV)
Figure 12. PWM Dimming, 2kHz, 50% Duty Cycle



Time (500 μs /DIV)
Figure 13. Startup through EN, $I_{LED} = 1.5A$



Time (10 μs /DIV)
Figure 14. Shutdown through EN, $I_{LED} = 1.5A$

Typical Waveforms (Continued)

Unless otherwise stated, the test conditions are applied: $V_{IN} = 24V$, LOAD = 2 LEDs in series, $F_S = 400kHz$, $L = 15\mu H$, $C_{OUT} = 10\mu F$, $T_A = 25^\circ C$.

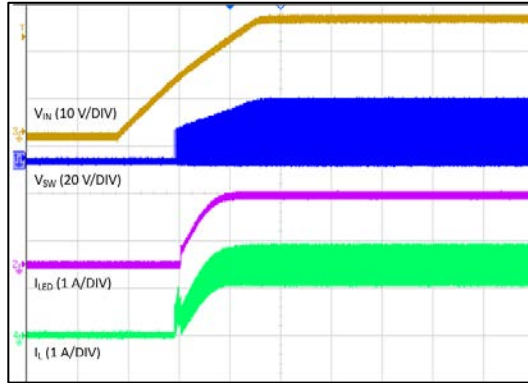


Figure 15. Startup through V_{IN} , $I_{LED} = 1.5A$

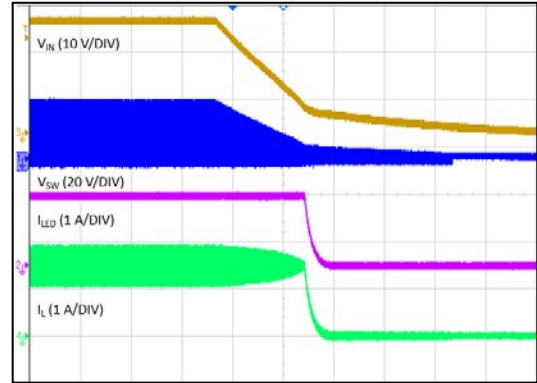


Figure 16. Shutdown through V_{IN} , $I_{LED} = 1.5A$

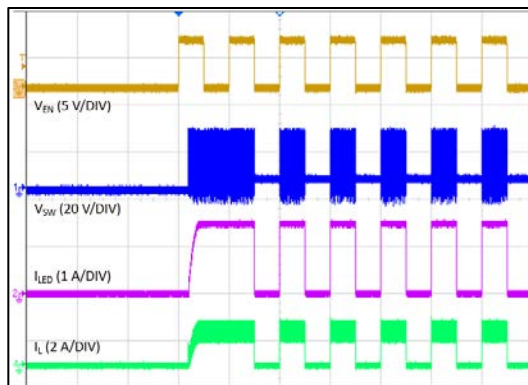


Figure 17. PWM Dimming Startup, 200Hz, 50% Duty Cycle, $I_{LED} = 1.5A$

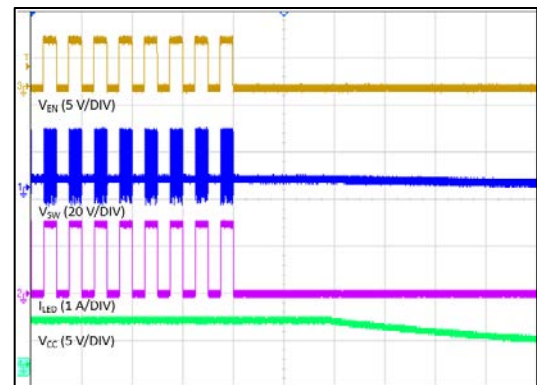


Figure 18. PWM Dimming Shutdown, 200Hz, 50% Duty Cycle, $I_{LED} = 1.5A$

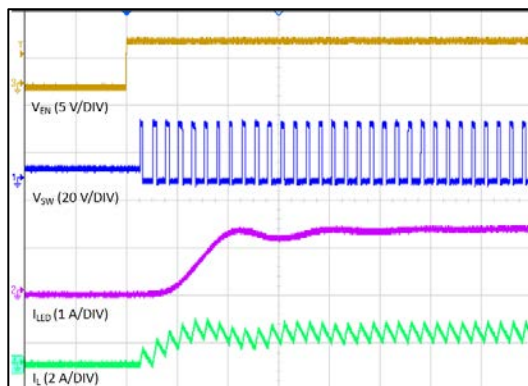


Figure 19. PWM Dimming Off – On Transient

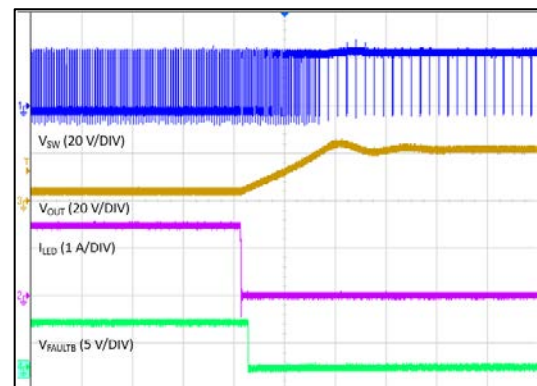
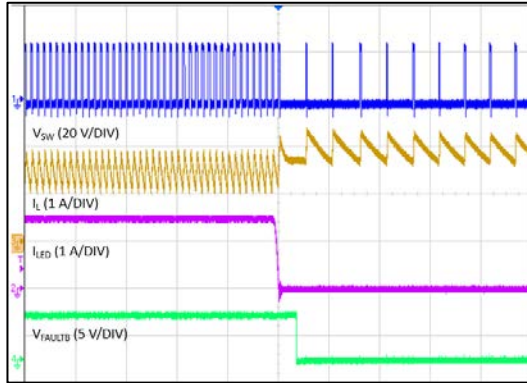


Figure 20. LED OPEN Entry

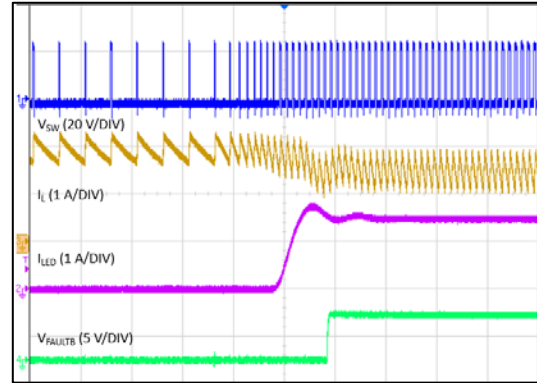
Typical Waveforms (Continued)

Unless otherwise stated, the test conditions are applied: $V_{IN} = 24V$, LOAD = 2 LEDs in series, $F_S = 400kHz$, $L = 15\mu H$, $C_{OUT} = 10\mu F$, $T_A = 25^\circ C$.



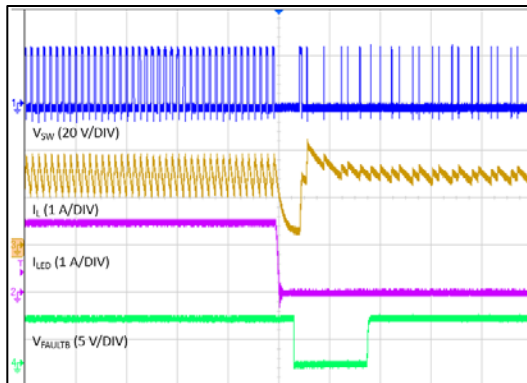
Time (20 μs /DIV)

Figure 21. LED+ Short to GND Entry



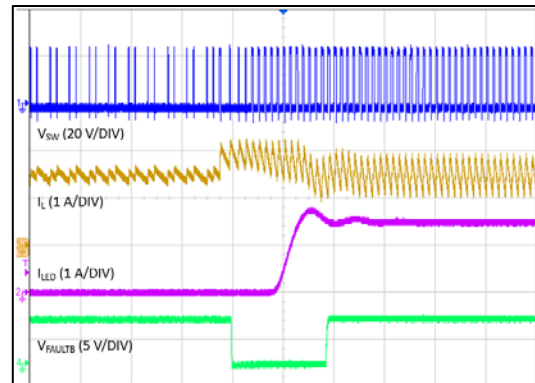
Time (20 μs /DIV)

Figure 22. LED+ Short to GND Recovery



Time (20 μs /DIV)

Figure 23. LED+ Short to LED- Entry



Time (20 μs /DIV)

Figure 24. LED+ Short to LED- Recovery

8 FUNCTIONAL DESCRIPTION

8.1 Overview

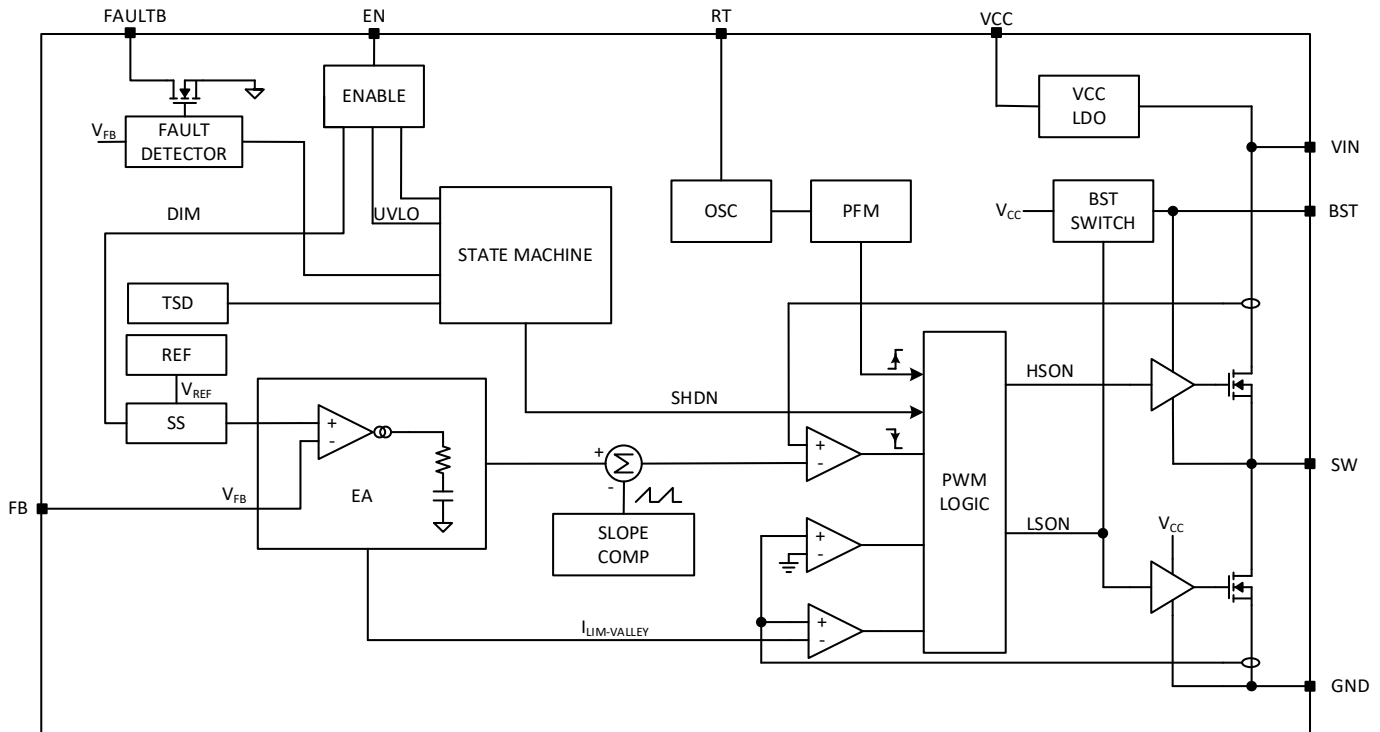
The LN33X61Q1 is a high efficiency, compact, synchronous step-down LED driver employing a constant frequency, peak current mode control architecture with internal compensation. It operates from an input voltage from 3.5V to 60V, and integrates both high-side and low-side power MOSFETs with up to 0.6A/1A/1.5A output current capabilities.

The LN33061Q1 has three switching frequency options: 400kHz, 1MHz and 2.1MHz. The LN33161Q1 with RT pin features an adjustable switching frequency through an external resistor connected between RT pin and GND pin. With a small minimum on-time, it enables a compact solution with small inductor and capacitor size and offers constant-frequency operation with high step-down ratio. Moreover, the LN33X61Q1 achieves the lowest possible dropout voltage with near 100% maximum duty cycle operation.

The LN33X61Q1 also features include programmable output current, internal soft start, and a wide array of protection functions such as cycle-by-cycle peak current limit, over temperature shutdown and recovery, and adjustable system UVLO.

The LN33X61Q1S offers a spread spectrum feature to optimize EMI performance. Emissions at the operating frequency and its sub-harmonics are mitigated by spreading the single-point operating frequency over an extended range.

8.2 Funtional Diagram



8.3 Functional Description Output Current Control and FB pin

The LN33X61Q1 employs a peak current mode PWM architecture. The average output current control is achieved by regulating the voltage on V_{FB} pin to an internal reference voltage.

To adjust the output current, connect a high-precision power resistor R_{FB} between LED and GND, and connect the common node of R_{FB} and LED to FB pin. The output current can be derived from:

$$I_{LED} = \frac{V_{FB}}{R_{FB}}$$

The steady state V_{FB} is typically 0.2V.

8.3.2 Internal LDO and V_{CC} Pin

V_{CC} powers the internal control circuits and the gate drivers for the internal power MOSFETs. V_{CC} must be decoupled to GND with a 1uF to 4.7uF ceramic capacitor.

8.3.3 V_{IN} UVLO and EN/DIM Pin

EN pin turns on and off the switching regulator and internal V_{CC} LDO. When the voltage on EN pin is above EN_{VCC-ON} , it turns on V_{CC} LDO.

When EN rises above a precise threshold, $EN_{VOUT-ON}$, it turns on the switching regulator. In order to achieve an accurate system V_{IN} UVLO function, an enable divider can be added between V_{IN} and GND. The switching regulator can thus be turned on at the programmable precise input voltage level, which is determined by:

$$V_{IN-RISING} = \left(1 + \frac{R_{ENT}}{R_{ENB}}\right) \times EN_{VOUT-ON}$$

8.3.4 PWM Dimming and EN/DIM Pin

PWM dimming is achieved by applying an external PWM signal at the EN/DIM pin. The chip is activated when logic high is applied to EN/DIM pin. When the EN/DIM pin is forced from high to low, the switching regulator will be turned off immediately, but the IC remains in standby mode for t_{EN-OFF} period, during which the IC can be restarted without delay and soft start for quick dimming response. The average LED current is proportionally scaled with the PWM duty cycle.

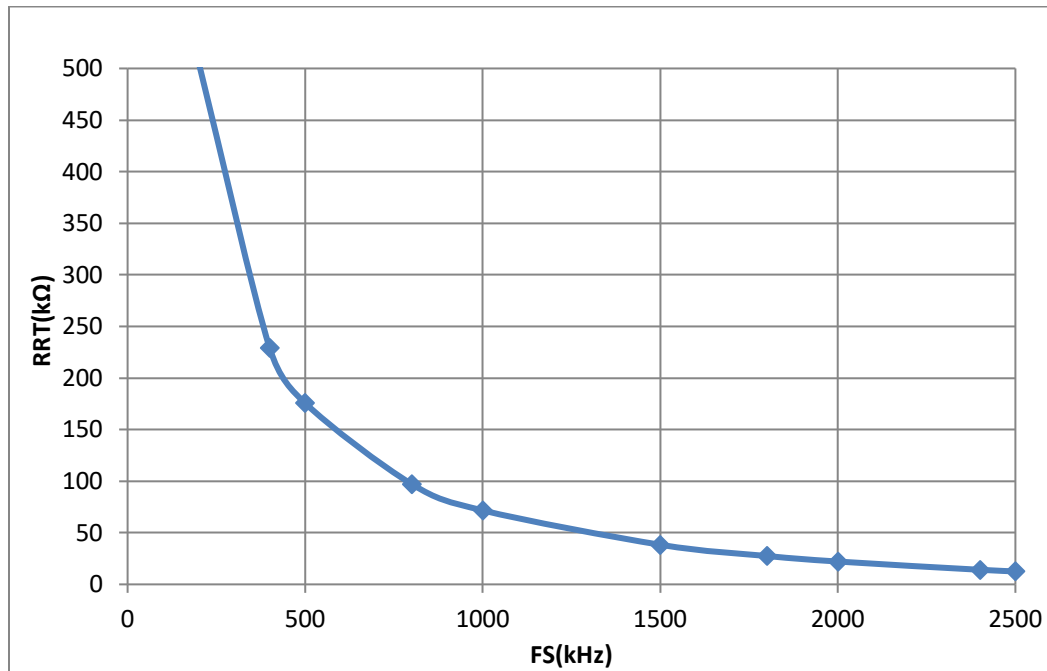
If the dimming signal is applied before starting up, the chip will initialize an **INTERNAL SOFT START** sequence, during this period the chip keeps switching and regulate at rated output current set by R_{FB} , regardless of the logic status of dimming signal. After soft start sequence is done, PWM dimming function will be activated.

8.3.5 Switching Frequency and RT Pin

The LN33061Q1 has three fixed frequency options: 400kHz, 1MHz and 2.1MHz. Meanwhile, the switching frequency of the LN33161Q1 with RT pin can be programmable between 200kHz to 2.5MHz by an external resistor connected between RT pin and GND pin. R_{RT} is calculated from:

$$R_{RT}(k\Omega) = \frac{152504}{F_s(kHz)^{1.069}} - 23$$

R_{RT} vs. Switching frequency is plotted as:



The following table lists some typical switching frequency and corresponding R_{RT} .

R_{RT} (kΩ)	F_s (kHz)
12.4	2500
22.1	2000
38.3	1500
71.5	1000
178	500
232	400
499	200

8.3.6 Fault Flag and FAULTB Pin

The FAULTB pin is connected to the open drain of an internal N-MOSFET. Externally, the FAULTB pin needs to be pulled up to a voltage source by a resistor. FAULTB goes high when the feedback voltage V_{FB} is within the fault window defined by fault over-voltage rising threshold and fault under-voltage falling threshold.

8.3.7 Over-Current Protection

8.3.7.1 Peak Current Protection

Peak current protection is a cycle-by-cycle protection inherited from the peak current mode control. The peak current is clamped to the peak current limit threshold.

8.3.7.2 Valley Current Protection

During the low side conduction, the low side power MOSFET current is sensed and compared to the valley current threshold. When the low side current is higher than the valley current threshold, the high side power MOSFET is not allowed to turn on for the next cycle.

8.3.8 Thermal Shutdown and Auto-Recovery

When the junction temperature exceeds 170°C, LN33X61Q1 shuts down the switching regulator to reduce thermal dissipation. It automatically restarts the switching regulator after the junction temperature drops back below 160 °C. The V_{CC} LDO regulator remains operational during over-temperature event.

8.3.9 Bootstrap Voltage, BST, and SW Pins

The internal gate driver for the high side Power MOSFET uses a bootstrapped supply from an external capacitor C_{BST} . C_{BST} connects between BST pin and SW. The voltage on C_{BST} is charged from V_{CC} through an internal switch when the low side power MOSFET conducts.

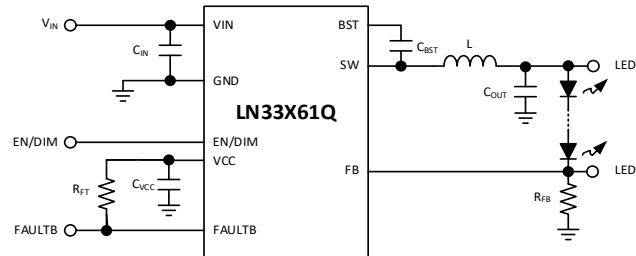
8.3.10 Low Drop-Out Operation

When input voltage is close to the output target voltage, LN33X61Q1 enters Low Drop-Out mode. The high-side MOSFET can be turned on for more than a switching period to maintain the output voltage regulation. When the input voltage is lower than the target voltage, the high-side power MOSFET is turned on for maximum allowable on time: t_{ON-MAX} . The low-side power MOSFET turns on briefly for $t_{OFF-MIN}$ to refresh the charge on C_{BST} .

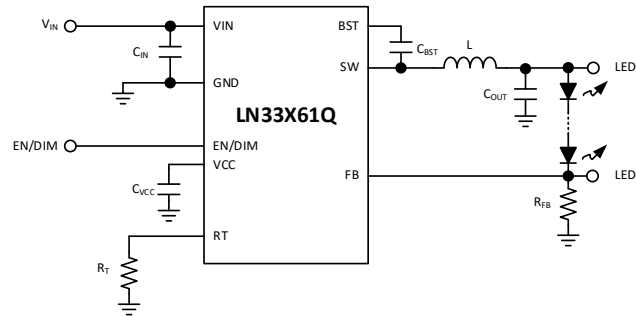
9 APPLICATION INFORMATON

9.1 Typical Applications

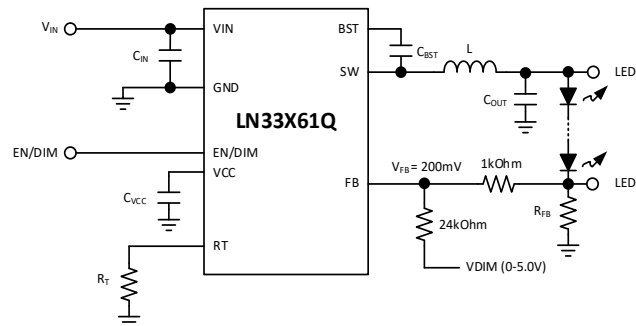
9.1.1 LN33061Q1 with FAULTB pin



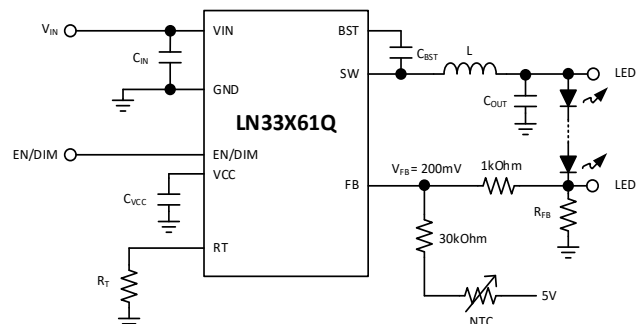
9.1.2 LN33161Q1 with RT pin



9.1.3 Analog dimming with an external voltage source



9.1.4 Thermal foldback of output current using NTC resistor



10 LAYOUT

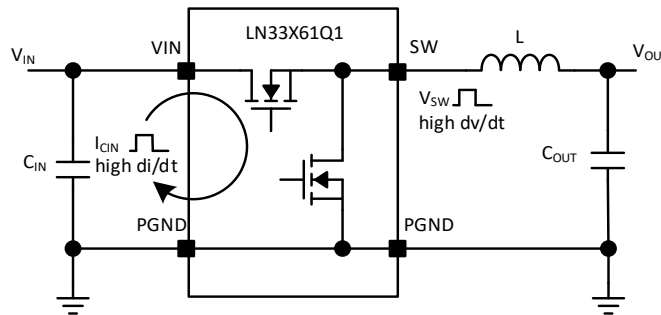
The performance of any switching converter depends as much upon the layout of the PCB as the component selection. The LN33X61Q1 is designed to meet the optimization requirements of PCB layout in the pin assignment. For example, VIN and GND pins are adjacent to each other, which is convenient for placing VIN bypass capacitors.

10.1 Layout Guidelines

Radiated EMI is generated by the high di/dt components in pulsing currents in switching converters. The larger area covered by the path of a pulsing current; the more electromagnetic emission is generated. As shown in the figure below, this part of the current flows from the VIN side of the input capacitors to high side switch, to the low side switch, and then returns to the ground of the input capacitors.

The key to minimize radiated EMI is to minimize the area of this pulsing current path, thus, placing high frequency ceramic bypass capacitor(s) as close as possible to the VIN and GND pins is necessary.

In addition, high dv/dt occurs on SW node during switching, so the trace between SW pin and inductor should be as short as possible, and just wide enough to carry the load current without excessive heating. Short and thick traces are highly recommended to minimize parasitic resistance. Besides, sensitive signal lines should be kept away from SW traces.

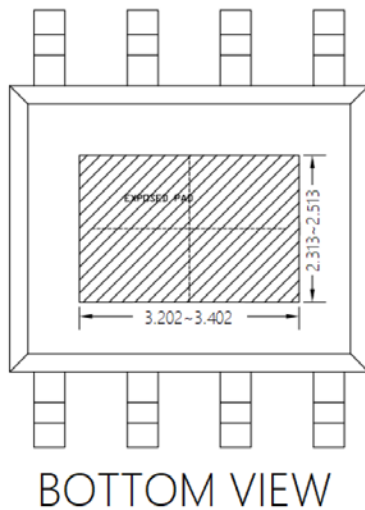
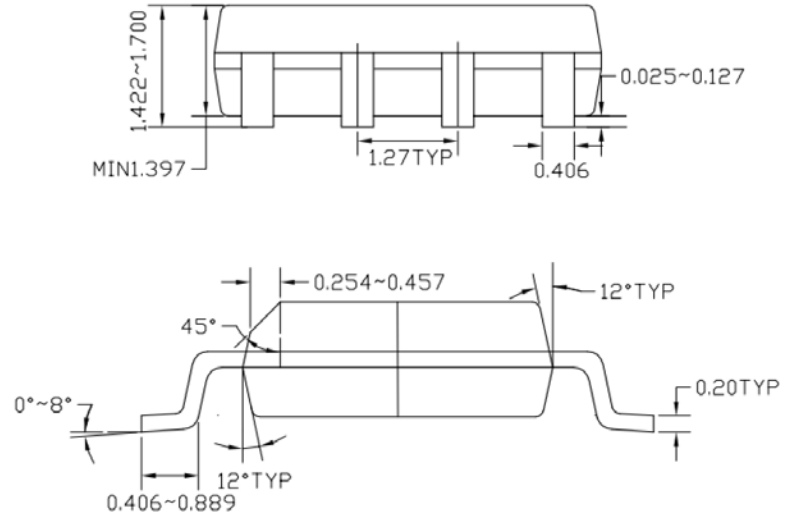
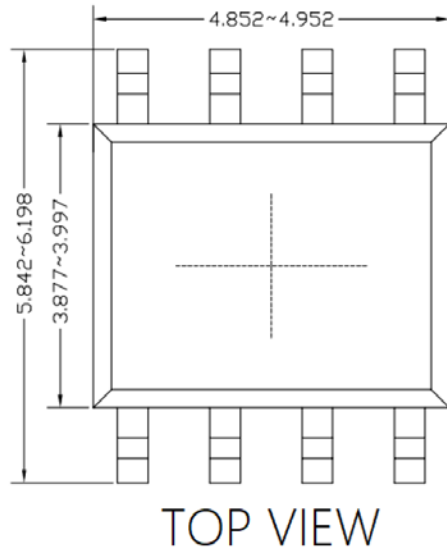


The following guidelines are provided to help users design a PCB with the best power conversion performance, thermal performance, and minimized generation of unwanted EMI.

1. Place high frequency ceramic bypass C_{IN} as close as possible to the LN33X61Q1 VIN and GND pins; a ceramic capacitor in small package (such as 0603) is still needed even if multiple input capacitors are implemented;
2. The high-current loop consisting of V_{IN} , V_{OUT} and GND should be as compact as possible;
3. The bypass capacitors of VCC should be arranged close to the pin, and return to the GND pin with the shortest connection;
4. It is recommended to use a four-layer board with 2oz top and bottom layers, and a dedicate ground plane on middle layer. Use a minimum 3 by 4 arrays of 10 mil thermal vias to connect the thermal pad of LN33X61Q1 to the system ground plane for heat sinking purpose;
5. The SW and BST nodes contains a lot of high-frequency noise, so the connection of the pins should be as short as possible, meanwhile, there should be sufficient width to conduct the current;
6. Sensitive analog signals, such as FB and RT, need to be far away from the noisy nodes, ground plane can be used as a shielding layer while routing these sensitive signals;
7. The feedback resistor of the FB connection must be located as close to the pin as possible;

11 PACKAGE INFORMATION

11.1 Package Outline

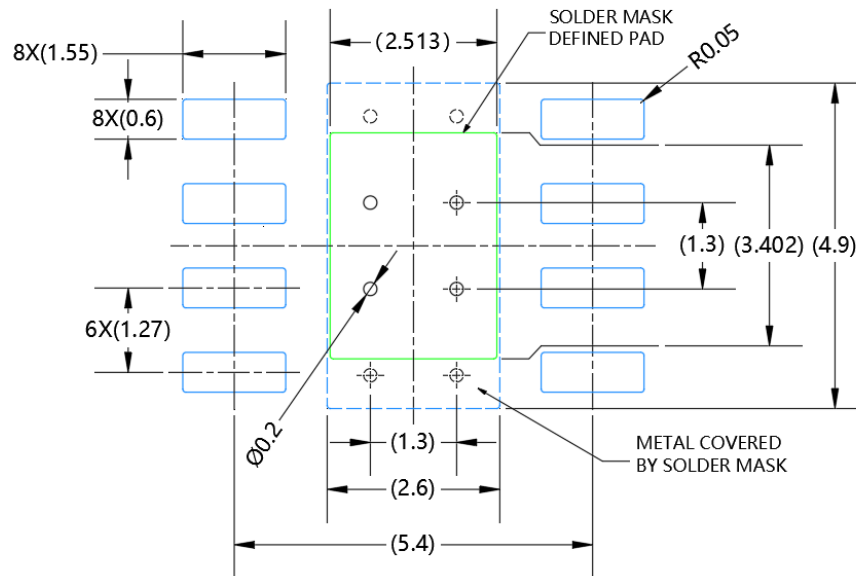


Notes:

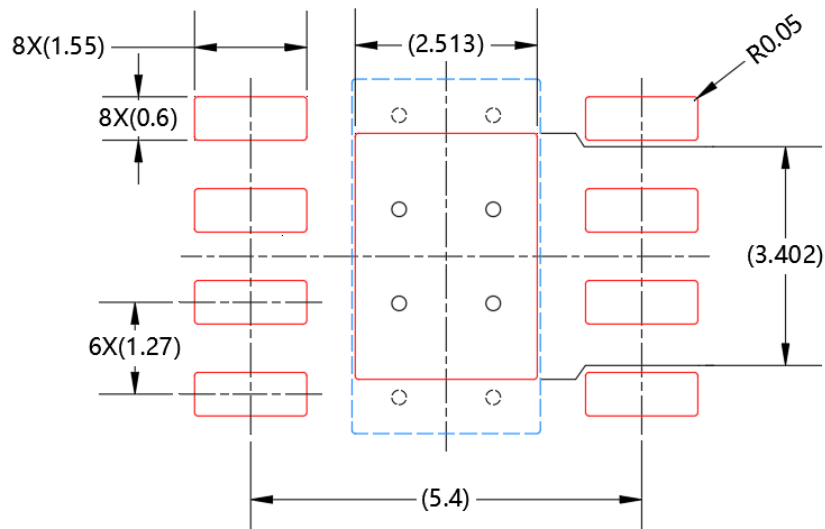
1. Both package length and width do not include mold flash.
2. Controlling dimension: mm
3. Reference JEDEC MS-013, MS-012
4. The size label of length and width in the drawing belong to the bottom size of the package.

11.2 Footprint Example

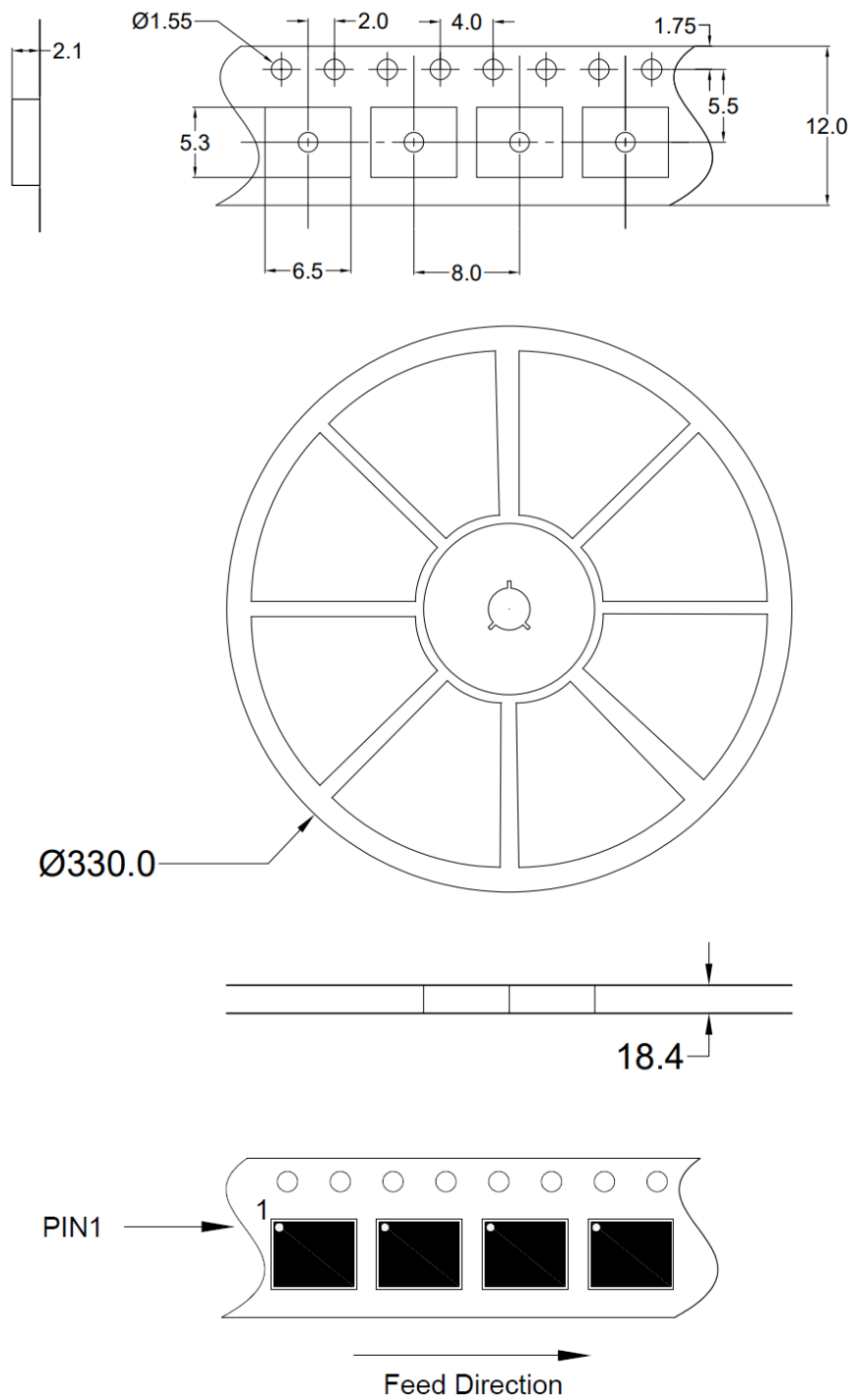
LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN



SOLDER PASTE EXAMPLE BASED ON 0.125 mm THICK STENCIL



12 TAPE AND REEL INFORMATION



Note:

1. All dimensions are in millimeters.

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